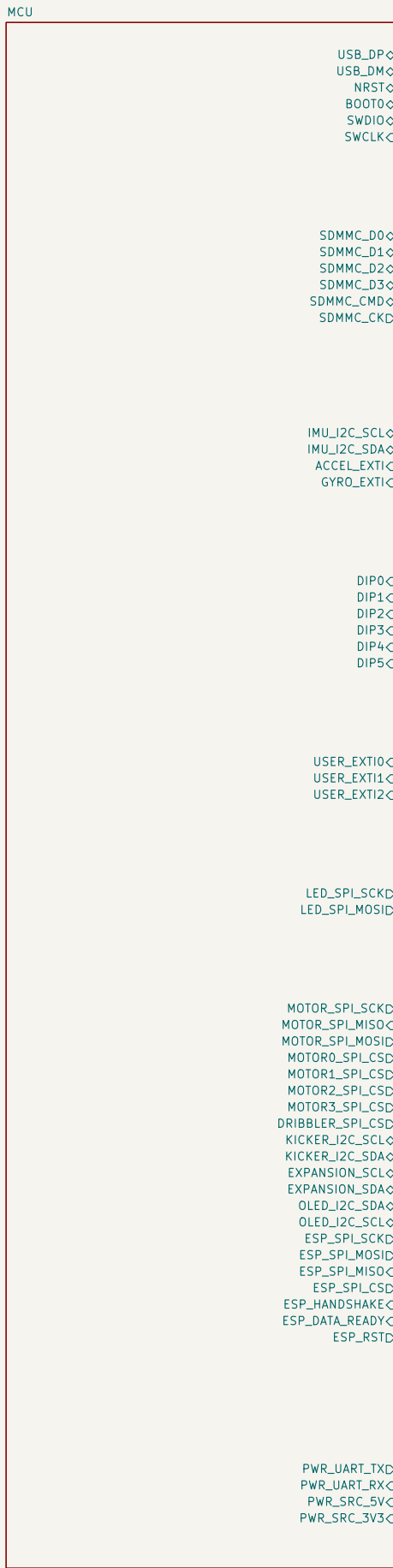
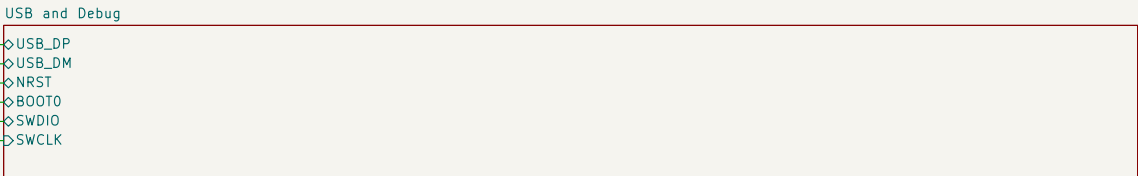
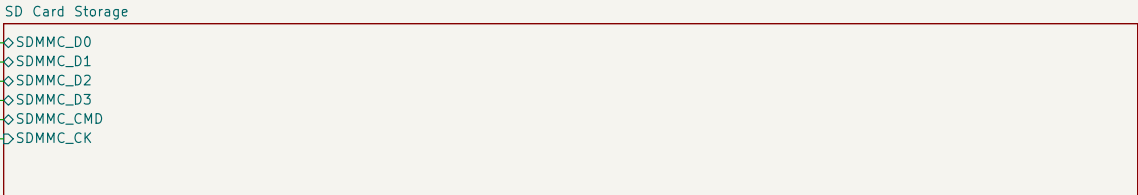




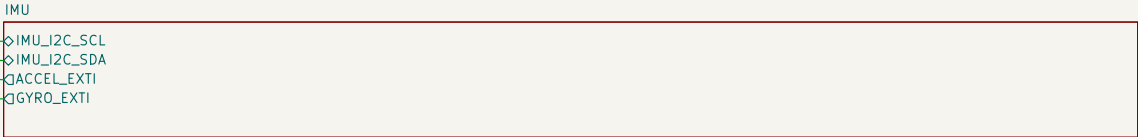
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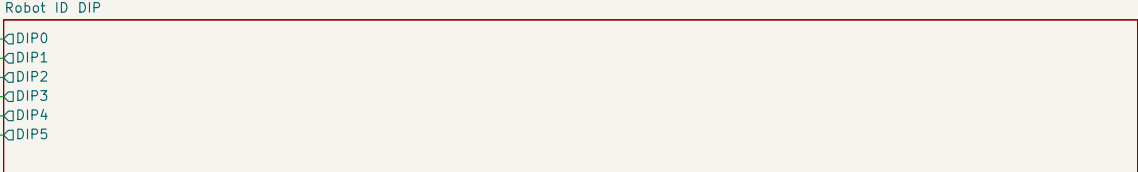
File: 03_USB_Debug.kicad_sch



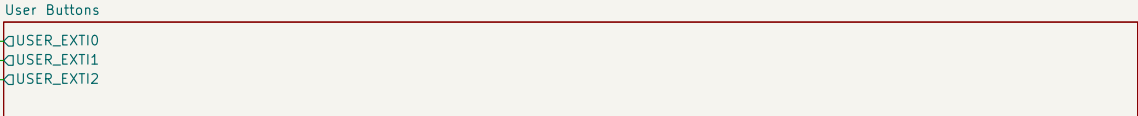
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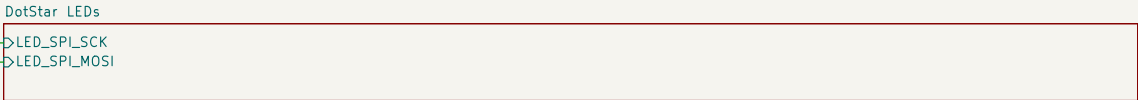
File: 05_IMU.kicad_sch



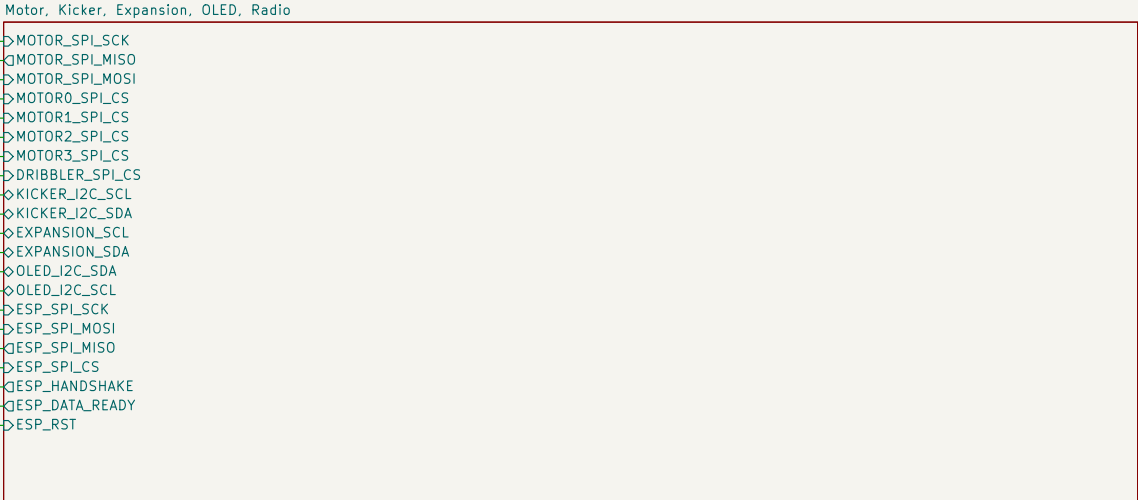
File: 08_DIP.kicad_sch



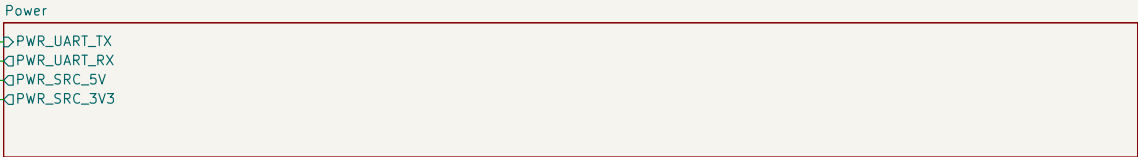
File: 09_Buttons.kicad_sch



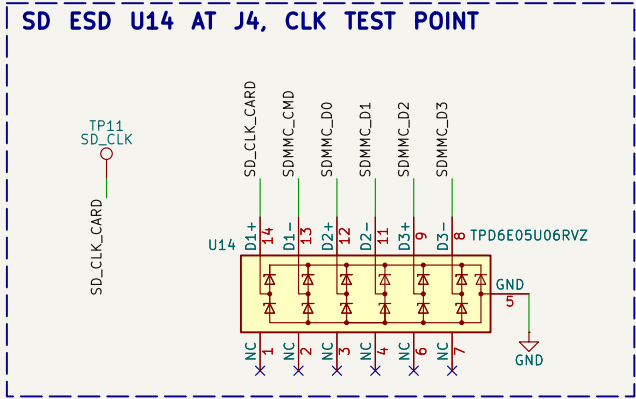
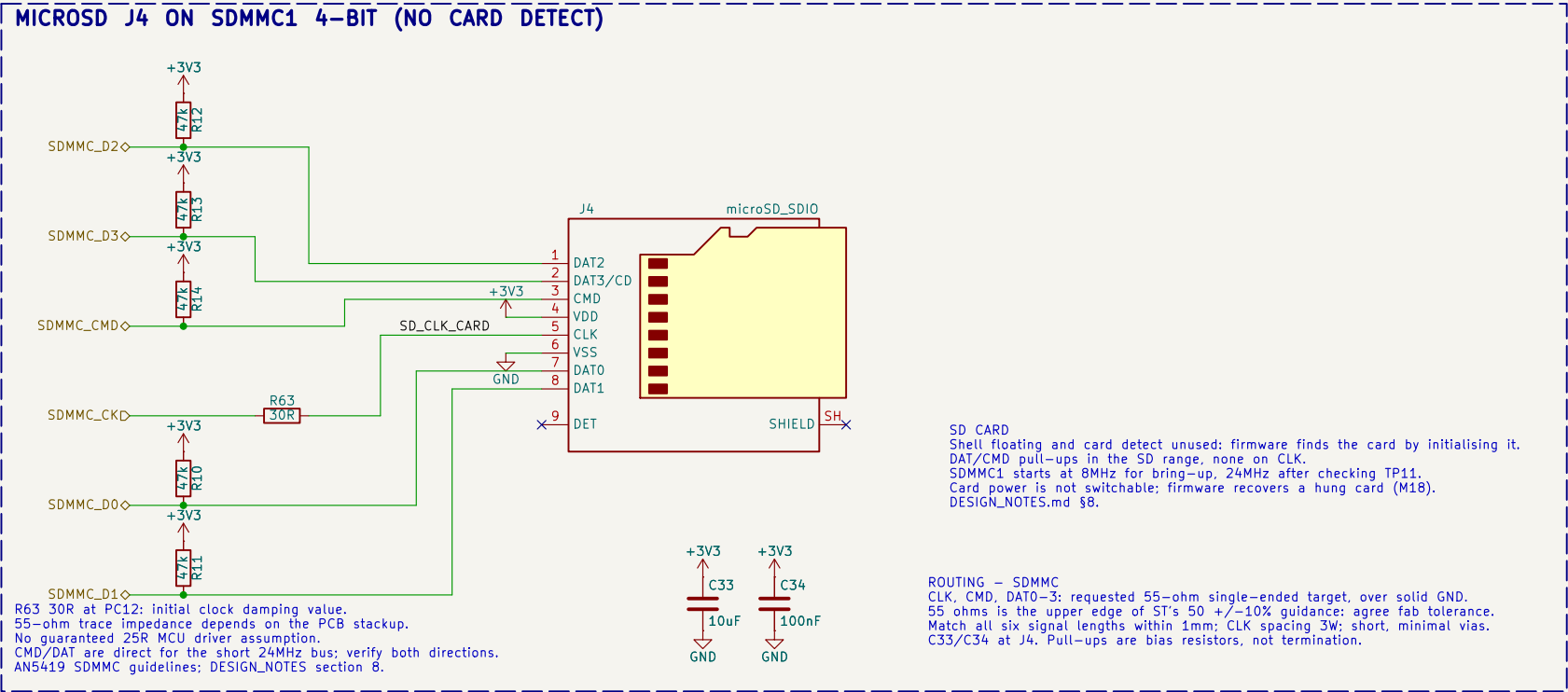
File: 06_LEDs.kicad_sch



File: 07_Connectors.kicad_sch

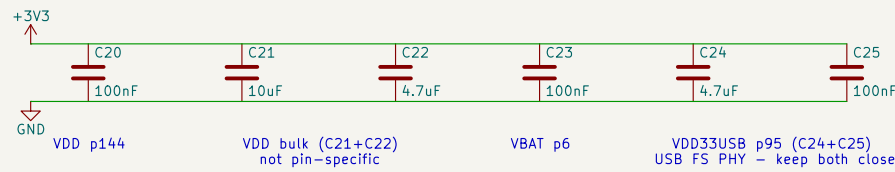
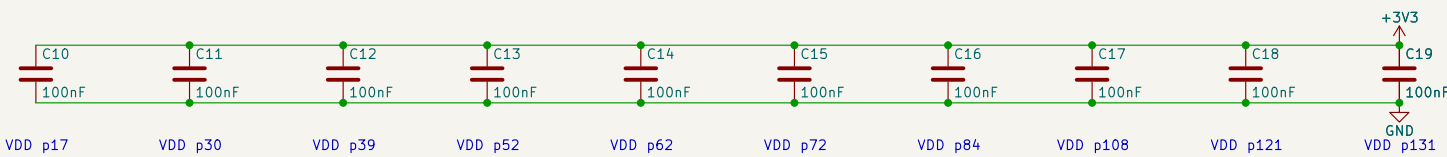


File: 01_Power.kicad_sch



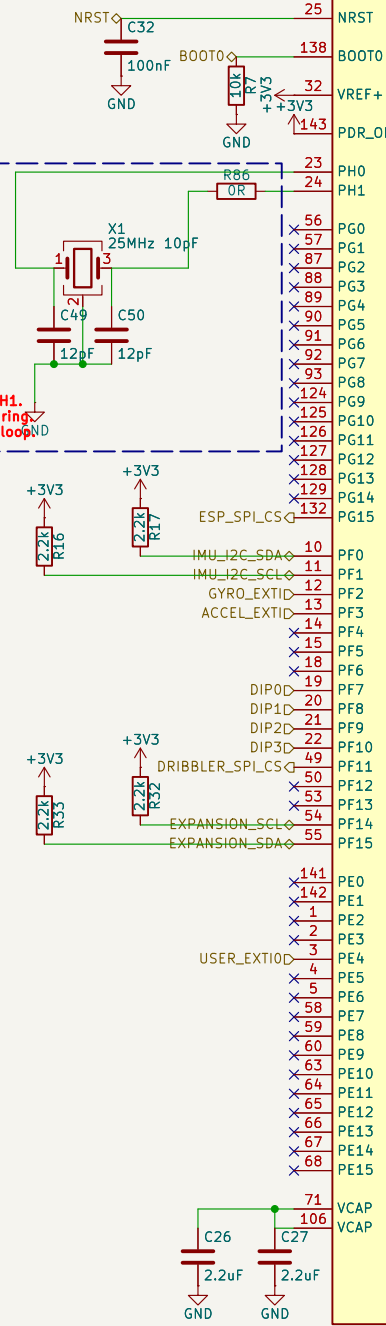
VDD DECOUPLING (EACH CAP AT ITS PIN)

ROUTING – DECOUPLING: each cap beside the STM32 pin named under it, via straight to GND.
C26/C27 (VCAP) tight to p71/p106. The netlist does not keep this pairing.



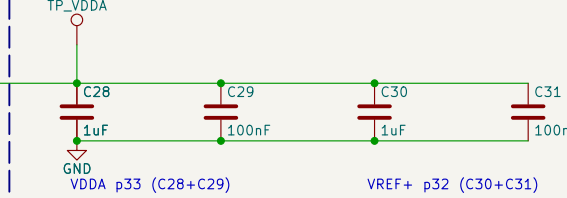
HSE 25MHZ CRYSTAL

ROUTING – HSE CRYSTAL
X1, C49, C50 within 5mm of PH0/PH1.
Short equal OSC traces, GND guard ring.
Nothing routed under or beside the loop.

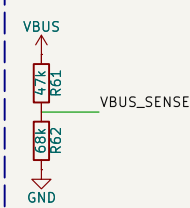


VDDA / VREF+ DECOUPLING

VDDA p33 + VREF+ p32 tie directly to +3V3
(no ferrite, as on the Nucleo). One cap pair
per pin. SEE/SGA/NOTES.md §6.

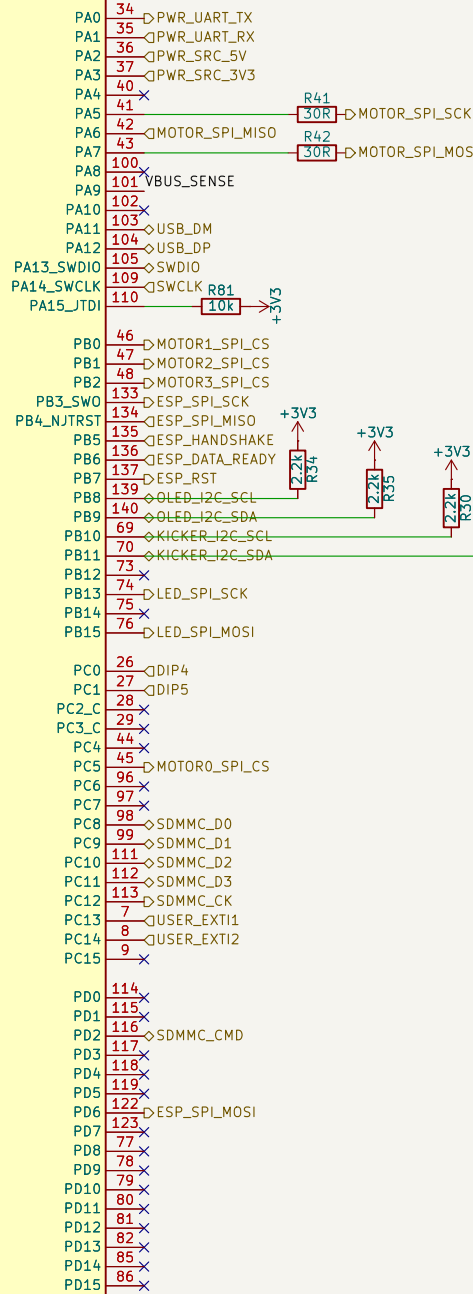


VBUS SENSE



R41/R42 30R: motor SPI SCK/MOSI damping, same initial value as R63.
55-ohm PCB traces do not imply an exact source match. MISO has none.
Qualify all five cable drops: DS13313 Tables 52/54; DESIGN_NOTES section 5.

ROUTING – MOTOR SPI (6 Mbit/s, 5 drops)
R41/R42 within 5mm of PA5/PA7, not at the connectors.



I2C PULL-UPS – all 2.2k, one pair per bus.
IMU R16/R17: kicker R30/R31; expansion R32/R33; OLED R34/R35.
TI SLV4889: 153pF maximum at 400kHz and +5% R.
OLED module 10k in parallel gives about 1.8k.
Measure loaded buses: no extra expansion-board pull-ups.

OPTIONAL I2C PULL-UP CHANGE (not applied – measure first)
R30/R31 and R32/R33: 2.2k -> 1.8k only if SCL rise time at TP16/TP17
exceeds 300ns with the real cable fitted. DESIGN_NOTES.md §6.

CLOCKING
X1 25MHz crystal: USB needs a tighter clock than the internal HSI48, and 25MHz starts reliably where a 48MHz crystal would not. C49/C50 match the crystal load. R86 (0R) leaves room for a drive-level resistor if measurement calls for one.
H CubeMX V0.3 assumes a 48MHz HSE: use the clock tree in DESIGN_NOTES.md §12 (C2, M16).
DIP4 = PC0 (p26), DIP5 = PC1 (p27); PF5/PF6 unused. No LSE: PC14 = USER_EXTI2.

SUPPLY PINS
LDO supply only. VDDA and VREF+ tie straight to +3V3, as on the Nucleo.
PDR_ON = +3V3 (internal reset). VBAT = +3V3 (no battery).
VCAP pins 71/106 are joined as the datasheet requires; KiCad ERC flags the tie (m21).
Details: DESIGN_NOTES.md §5.

ROUTING - USB FS
 USB_DP/USB_DM: 90R differential pair, length-matched within 1mm,
 over solid GND, no vias if possible, no stubs.
 Join J2 A6/B6 and A7/B7 at the connector. U5 right at J2, then to PA11,
 R8/R9 near J2. SWD to J3 short.

SHIELD
 SH A1
 GND

TVS0500DRV D12
 Drv Det &
 F1
 MF-MSMF075/16X
 R8
 R9
 CC1
 CC2
 A5
 B5
 CC1
 CC2
 A7
 B7
 D-
 D-
 D+
 D+
 A8
 B8
 SBU1
 SBU2
 GND

PWR_FLAG
 TP3 TP-VBUS
 VBUS

USB-C-DFU
 J2
 VBUS

USB_DP
 USB_DM
 +3V3
 U5
 USBLC6-25C6
 GND

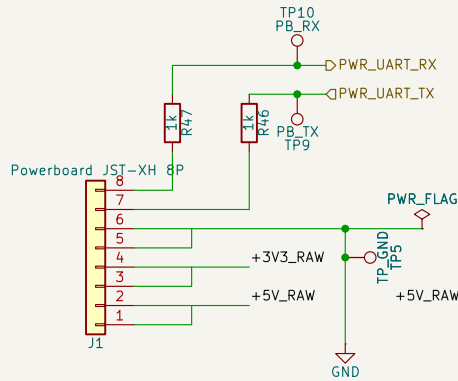
USB POWER LIMIT: F1 holds 0.55A at 50C.
~0.85A concurrent base + Wi-Fi + LEDs exceeds this.
Use powerboard for full operation; USB bench mode restricted.
A larger fuse alone does not increase allowed USB current.

SWD DEBUG J3, RESET AND BOOT BUTTONS

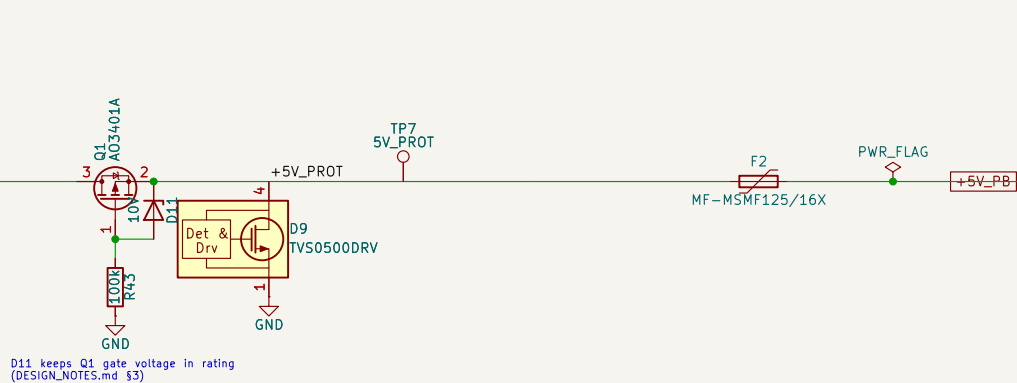
SWD: R39/R40 removed; direct SWCLK/SWDIO connections.
 Use short GND-referenced probe wiring; start SWCLK at 1MHz.
 Series damping is optional SI tuning, not required DC matching.
 NRST: internal 30–50k pull-up + C32 100nF (DS13313 section 6.3.17).
 J3 must be keyed. U5 rail pin stays on +3V3.

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J3 must be keyed. U5 rail pin stays on +3V3.

POWERBOARD CONNECTOR J1



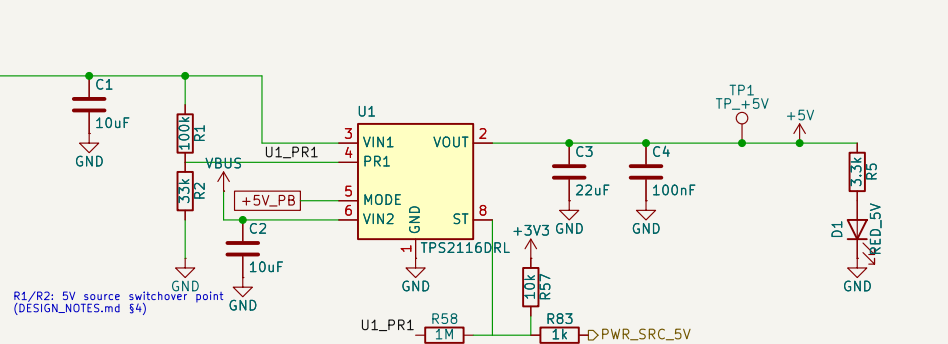
5V INPUT: Q1 REVERSE POLARITY, D9 TVS, F2 FUSE



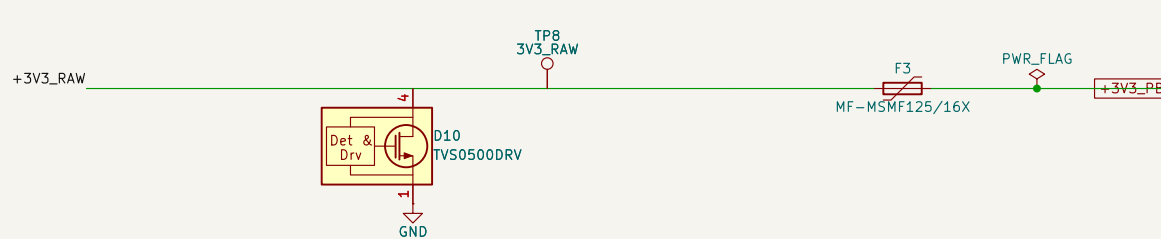
ROUTING - POWER INPUT

J1 -> Q1/D9/D10 -> F2/F3 -> U1/U2: 2A traces, >=1.0mm on 1oz outer (or pours).
Q1, D9 and D10 right at J1 (D9 stays after Q1). Short wide GND via at each TVS.
C1/C2/C7 at TPS2116 VIN, C3/C8 at VOUT.
U3 tab to a copper pour. R46/R47 near J1, UART pair away from the power pours.

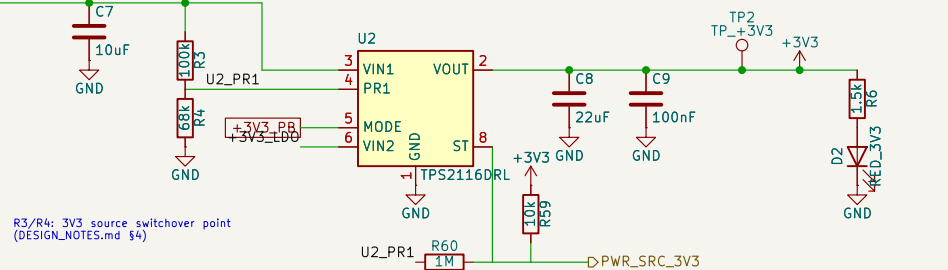
5V MUX U1: POWERBOARD, ELSE USB



3V3 INPUT: D10 TVS, F3 FUSE



3V3 MUX U2: POWERBOARD, ELSE USB LDO



USB VBUS AND 3V3 LDO U3



USB VBUS INPUT
F1 PTC fuse (sheet 03) opens on a cable or board short. D12 at J2 absorbs surges.
C55 meets the USB minimum VBUS capacitance. No inrush limiter.
Details: DESIGN_NOTES.md §4.

LIMITS

No overvoltage cut-off on the powerboard rails: the powerboard must regulate (M1, M2).
The TVS diodes only take surges. PTC fuses are slow: they protect wiring and connectors.
Details: DESIGN_NOTES.md §3 and §13.

POWER MUX - U1/U2 TPS2116, priority mode
Powerboard (VIN1) is used whenever present; USB (VIN2) is the bench fallback.
No output diodes needed: the TPS2116 blocks reverse current into an unused input.
ST tells the MCU which source is active (PWR_SRC_5V/3V3). R83 protects PA2 from the bootloader.
The radio (J14, +3V3) and DotStars (+5V) run from the mux outputs, so they also draw from USB.
Details: DESIGN_NOTES.md §2 and §4. Sag at switchover with USB attached: M7.

J1 POWERBOARD CONNECTOR - JST-XH 8P
1-2 +5V 3-4 +3V3 5-6 GND 7 PWR_UART_TX 8 PWR_UART_RX (order to confirm with powerboard)
Q1 blocks a reversed 5V input. D9/D10 absorb surges.
F2/F3 PTC fuses open on a sustained short. No undervoltage or overvoltage cut-off (kept simple).
Why these parts and values: DESIGN_NOTES.md §3. Open issues M1-M3.

Sheet: /Power/

File: 01_Power.kicad_sch

Title:

Size: A3

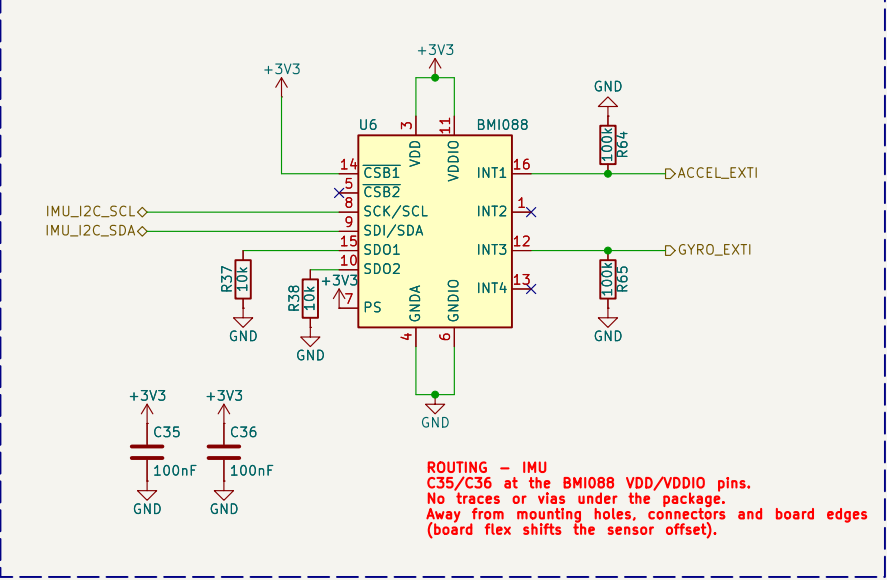
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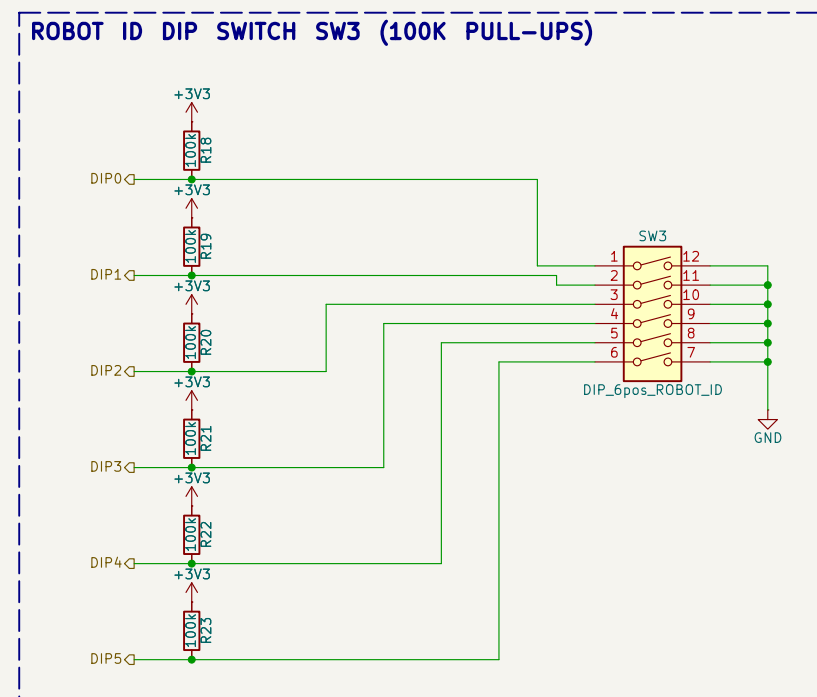
Rev:

KiCad E.D.A. 10.0.0

Id: 6/10

IMU U6 BMI088: I2C, ACCEL AND GYRO INTERRUPTS





DIP PULL-UPS = 100k
Solid logic high with little current through each closed switch.
Read at boot only; add 100nF per pin if read continuously.
DIP4 = PC0, DIP5 = PC1 (V0.3). DESIGN_NOTES.md §9.

Sheet: /Robot ID DIP/
File: 08_DIP.kicad_sch

Title:

Size: A3

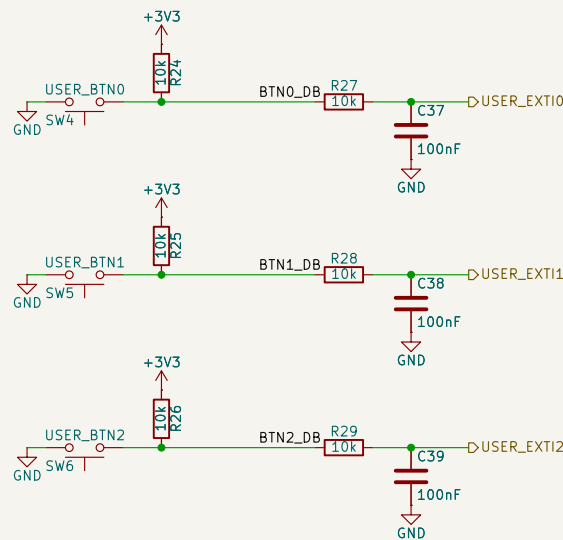
Date:

Rev:

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KiCad E.D.A. 10.0.0	

Id: 9/10

USER BUTTONS SW4-SW6 WITH RC DEBOUNCE



BUTTON DEBOUNCE (same for SW4, SW5, SW6)
Pull-up R24/R25/R26 = 10k, series R27/R28/R29 = 10k, C37/C38/C39 = 100nF, no inductor.
Press (C discharges through series R): $\tau = R27 \times C37 = 10k \times 100nF = 1.0ms$
Release (C charges through pull-up + series R): $\tau = (R24 + R27) \times C37 = 20k \times 100nF = 2.0ms$
With 1% resistors and 10% caps: press 0.9–1.1ms, release 1.8–2.2ms. Settles in $\sim 5 \tau$.
The RC removes sub-millisecond bounce spikes; firmware ignores re-triggers for $\sim 10ms$.
The series R also limits current into the MCU pin from the discharging cap.
DESIGN_NOTES.md §9.